Numerical Analysis

Topic 2:

Solution of Nonlinear Equations

Sections 5-11:

Lecture 4

False Position Method

False Position Method

Motivations

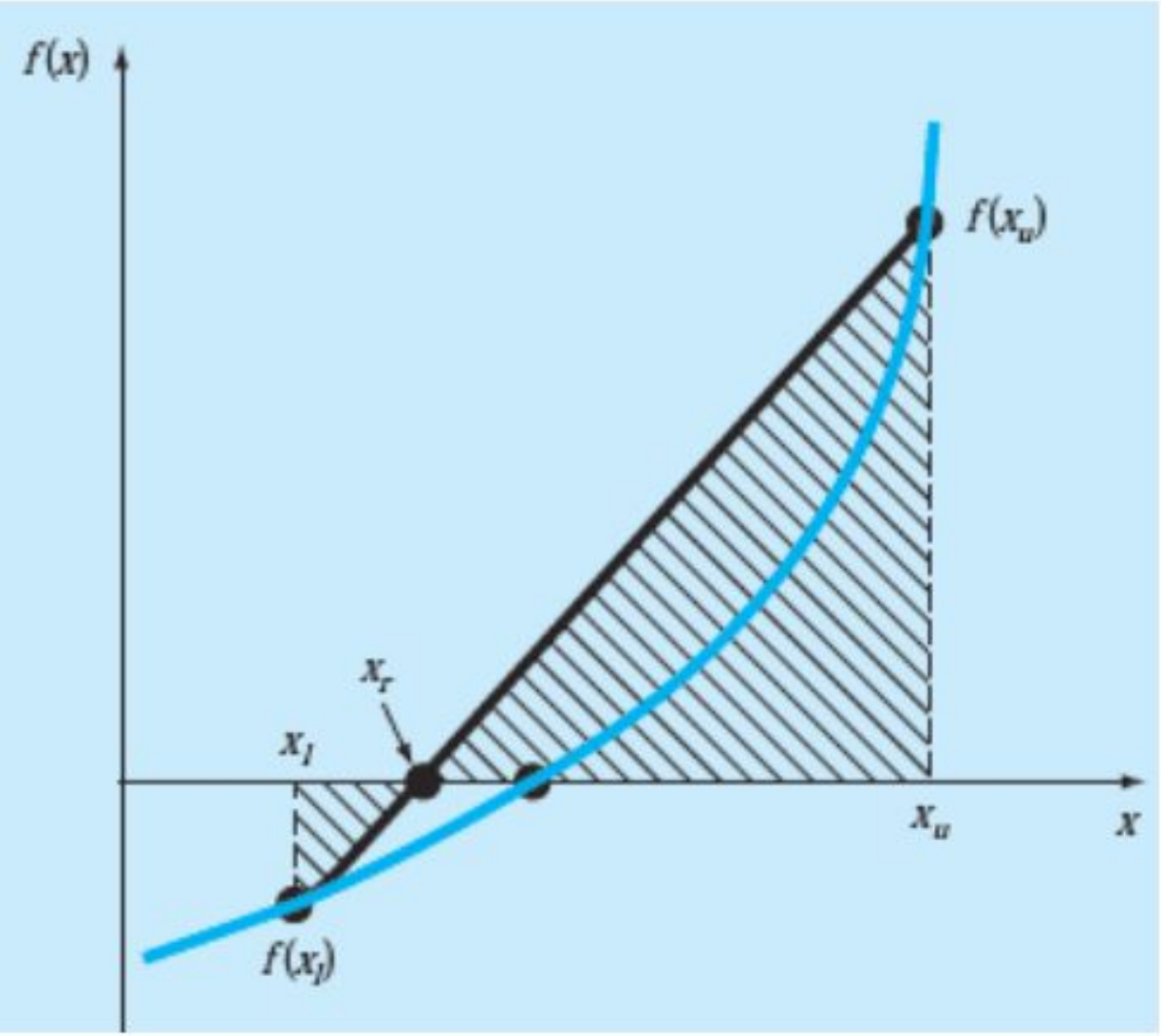
- Although bisection is a perfectly valid technique for determining roots, false position is an alternative based on a graphical insight.
- A shortcoming of the bisection method is that, in dividing the interval from x_a to x_b into equal halves, **no account is taken of the magnitudes of $f(x_a)$ and $f(x_b)$.**
- For example, if $f(x_a)$ is much closer to zero than $f(x_b)$, it is likely that the root is closer to x_a than to x_b .

False Position Method

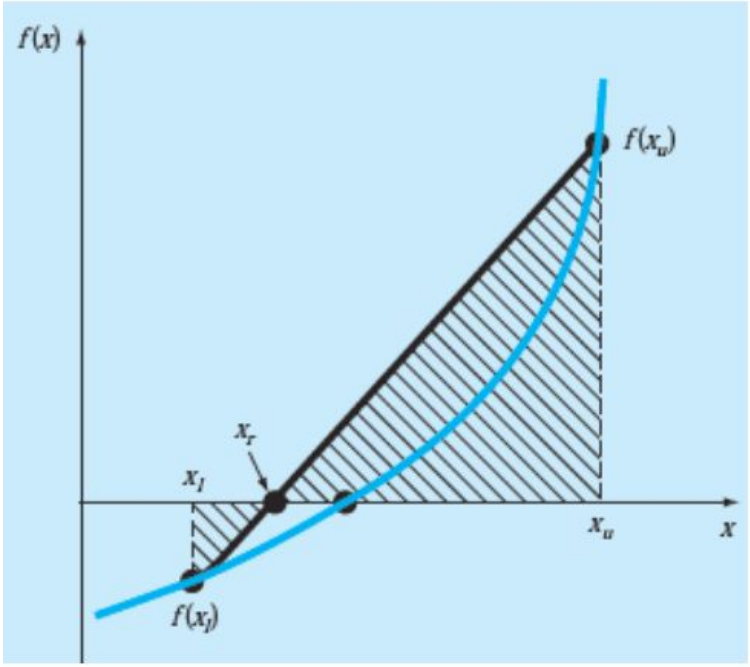
Motivations

- An alternative method that exploits this graphical insight is to **join $f(x_a)$ and $f(x_b)$ by a straight line.**
- The **intersection** of this line with the ***x* axis** represents an improved **estimate of the root.**

False Position Method



False Position Method



$$\frac{f(x_L)}{(x_r - x_L)} = \frac{f(x_u)}{(x_U - x_r)}$$

$$x_r = x_u - \frac{f(x_u)(x_L - x_U)}{f(x_L) - f(x_U)}$$

False Position Method Algorithm

Step 1: Choose Initial Interval

Select a and b such that:

$$f(a)f(b) < 0$$

(There must be a sign change.)

Step 2: Compute New Approximation

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

False Position Method Algorithm

Step 3: Test Sign

- If $f(a)f(c) < 0$

  → root lies in $[a, c]$

  → set $b = c$
- Else if $f(b)f(c) < 0$

  → root lies in $[c, b]$

  → set $a = c$

Step 4: Stopping Criteria

Stop if one of the following is satisfied:

- $|f(c)| < \varepsilon$
- $|c - c_{\text{previous}}| < \varepsilon$
- Maximum iterations reached
- Absolute Error |b-a|/2< ε

Step 5: Repeat

Return to Step 2.

False Position Method

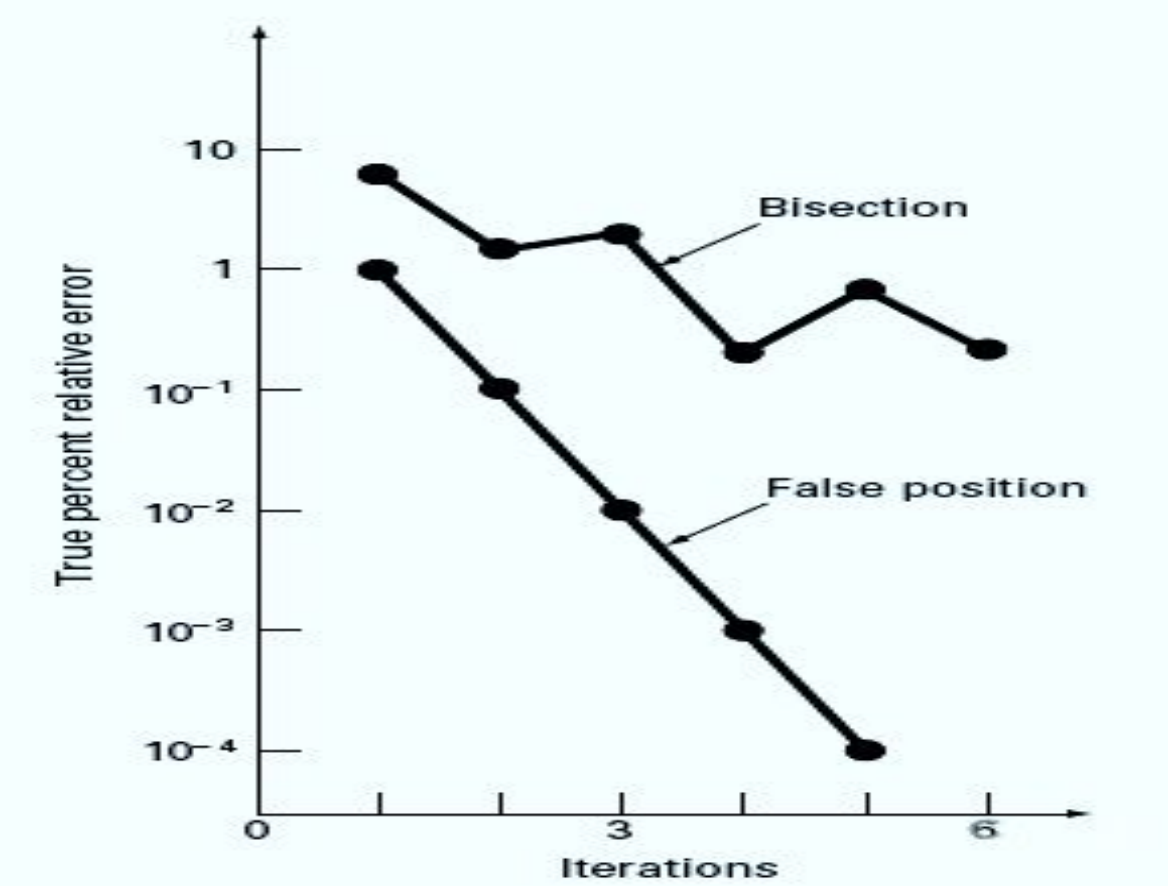
- Example

Use the false-position method to determine the root of the equation(the initial interval= [12,16])

$$f(c) = \frac{9.8(68.1)}{c}\left(1 - e^{-(c/68.1)10}\right) - 40$$

False Position Method

- In many cases False position converges faster



Example

Use False-position method to find the roots of the function

$f(x) = x^3 + 2x^2 - 3x - 1$

take $[x_L, x_U] = [1, 2]$. The approximate error should be less than 10^{-3}

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Function f is continuous on [1, 2] and $f(1) = -1, f(2) = 9$ so there is a root in [1, 2].

$$x_m = x_u - \frac{f(x_u)(x_L - x_U)}{f(x_L) - f(x_U)}$$

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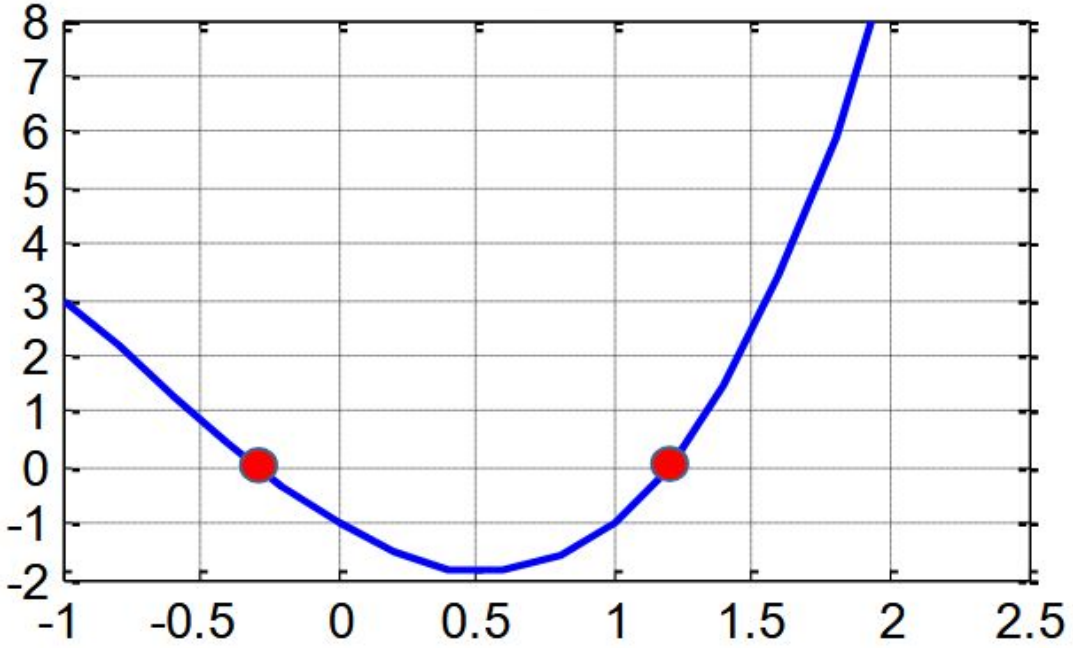
$$x_m = 2 - \frac{9 \times (1 - 2)}{-1 - 9} = 1.1$$

- $f(x_L) = f(1) = -1, f(2) = 9, \quad f(1.1) = -0.549$
- $f(x_u)f(x_m) < 0$ ➔ then $x_L = x_m$ and $x_u = x_u$

Solution of example using False-position method

Iteration no.	x_L	x_u	x_m	$\lvert\epsilon_a\rvert$
1	1	2	1.1	-
2	1.10	2	1.1517	4.4926
3	1.1517	2	1.1768	2.1326
4	1.1768	2	1.1886	0.9916
5	1.1886	2	1.1941	0.4565
6	1.1941	2	1.1966	0.2092
11	1.1987	2	1.1987	8.6600e-004

- From the figure we can see that the function has two solutions. One is between [-0.5, 0] and the second is between [1, 1.5].
- The False-position method gives the second solution which is 1.1987 after 11 iteration.



Solution of example using Bisection method

Iteration no.	x_m	$\lvert\epsilon_a\rvert$
1	1.5000	0.333
2	1.2500	0.200
3	1.1250	0.111
4	1.1875	0.0526
5	1.2188	0.0256
6	1.2031	0.0130
7	1.1953	0.0065
8	1.1992	0.0033
9	1.1973	0.0016
10	1.1982	8.150e-4
11	1.1987	4.073-4

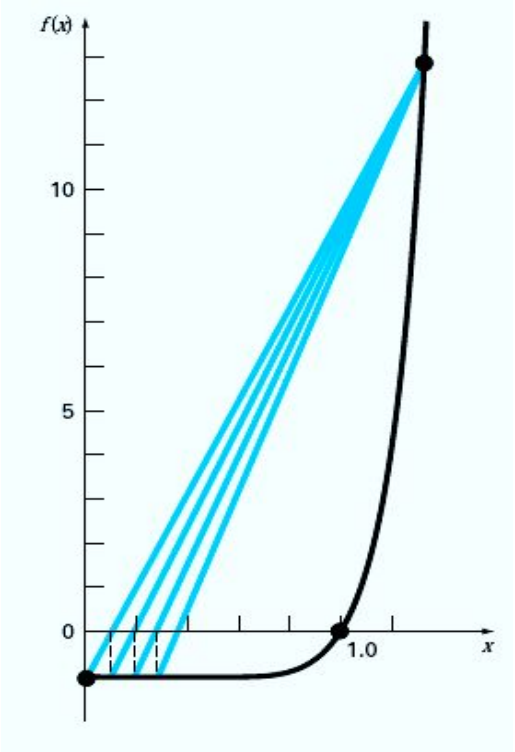
Pitfalls of the False-Position Method

The convergence of the algorithm is very slow for some cases

False Position Method

- Pitfalls of False position method:
  - Consider $f(x) = x^{10} - 1$ between $x = 0$ and $x = 13$

The convergence of the algorithm is very slow for False-Position Method in this case



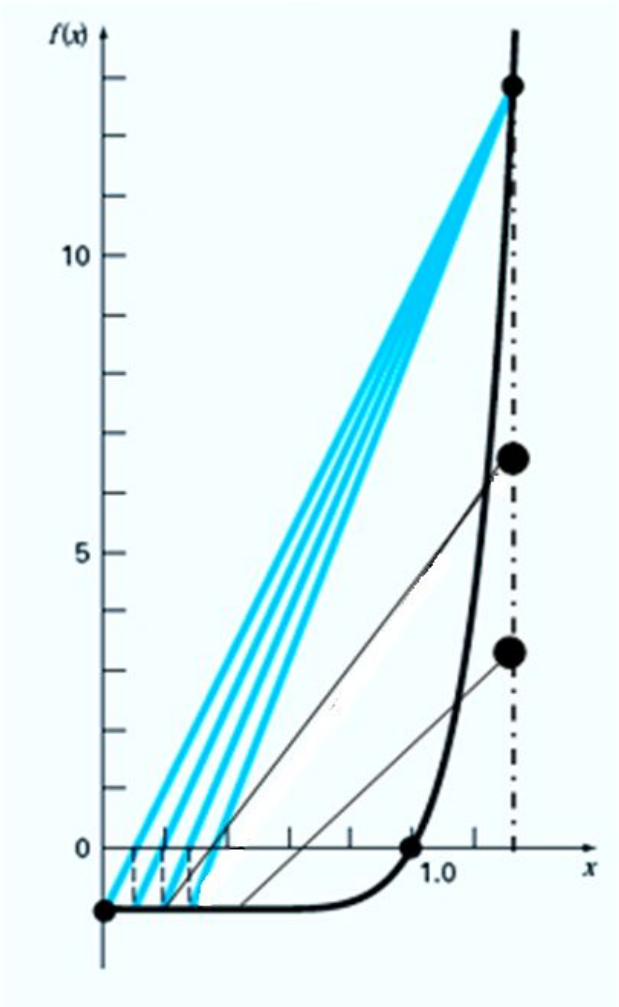
Modified False Position Method

1. Compute:

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

2. If $f(a)f(c) < 0$:
   - $b = c$
   - If b did not move in previous step → set $f(b) = f(b)/2$
3. Else:
   - $a = c$
   - If a did not move in previous step → set $f(a) = f(a)/2$
4. Repeat until tolerance satisfied.

Modified False Position Method



Problem Statement. Use bisection and false position to locate the root of

$$f(x) = x^{10} - 1$$

between $x = 0$ and 1.3.

Solution. Using bisection, the results can be summarized as

BISECTION METHOD

Iter.	a	b	c	f(c)	Absolute Error	Relative Approx. Error (%)
1	0.000000	1.300000	0.650000	-0.986537	0.350000	-
2	0.650000	1.300000	0.975000	-0.223670	0.025000	33.3333
3	0.975000	1.300000	1.137500	2.625207	0.137500	14.2857
4	0.975000	1.137500	1.056250	0.728889	0.056250	7.6923
5	0.975000	1.056250	1.015625	0.167888	0.015625	4.0000
6	0.975000	1.015625	0.995313	-0.045886	0.004687	2.0408
7	0.995313	1.015625	1.005469	0.056077	0.005469	1.0101
8	0.995313	1.005469	1.000391	0.003913	0.000391	0.5076
9	0.995313	1.000391	0.997852	-0.021270	0.002148	0.2545
10	0.997852	1.000391	0.999121	-0.008756	0.000879	0.1271
11	0.999121	1.000391	0.999756	-0.002438	0.000244	0.0635
12	0.999756	1.000391	1.000073	0.000733	0.000073	0.0317

The standard false-position method

Iter.	a	b	c	f(c)	Absolute Error	Relative Approx. Error (%)
1	0.000000	1.300000	0.094302	-1.000000	0.905698	-
2	0.094302	1.300000	0.181861	-1.000000	0.818139	48.14
3	0.181861	1.300000	0.263305	-0.999998	0.736695	30.93
4	0.263305	1.300000	0.339062	-0.999980	0.660938	22.35
5	0.339062	1.300000	0.409537	-0.999869	0.590463	17.21
6	0.409537	1.300000	0.475119	-0.999411	0.524881	13.80
7	0.475119	1.300000	0.536148	-0.997964	0.463852	11.38
8	0.536148	1.300000	0.592933	-0.994718	0.407067	9.58
9	0.592933	1.300000	0.645748	-0.988664	0.354252	8.18
10	0.645748	1.300000	0.694840	-0.978748	0.305160	7.07
11	0.694840	1.300000	0.740429	-0.963956	0.259571	6.16
12	0.740429	1.300000	0.782716	-0.943414	0.217284	5.40

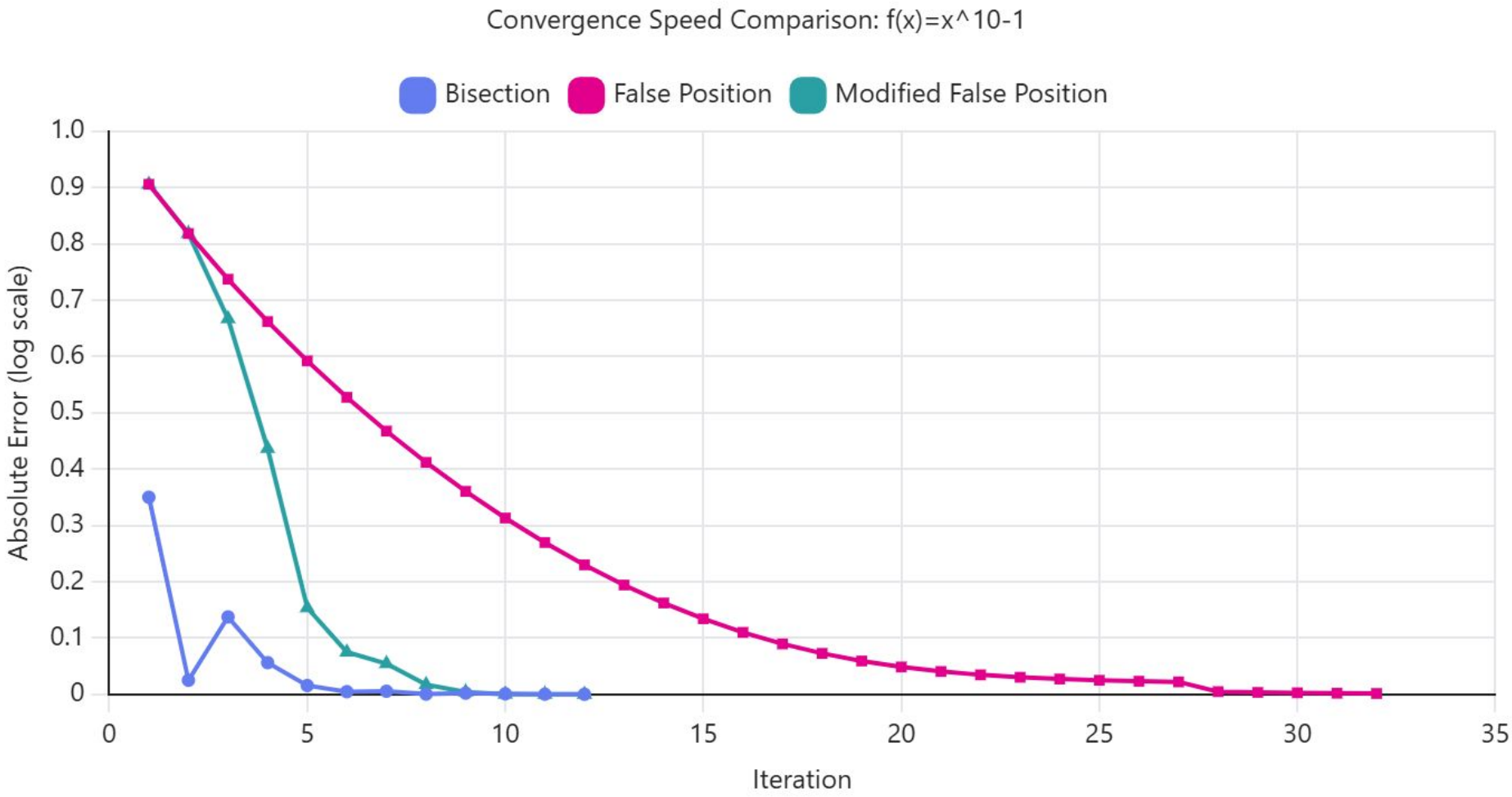
The modified false position method

Iter	a	b	f(a)	f(b)	c	f(c)	Abs Error	Rel Approx %
1	0.000000	1.300000	-1.000000	12.785849	0.094300	-1.000000	0.905700	-
2	0.094300	1.300000	-1.000000	6.392925	0.181759	-1.000000	0.818241	48.1183
3	0.181759	1.300000	-1.000000	3.196462	0.333017	-0.999983	0.666983	45.4206
4	0.333017	1.300000	-0.999983	1.598231	0.563442	-0.996775	0.436558	40.8960
5	0.563442	1.300000	-0.996775	0.799116	0.846364	-0.811388	0.153636	33.4279
6	0.846364	1.300000	-0.811388	0.399558	1.074910	1.059310	0.074910	21.2619
7	0.846364	1.074910	-0.811388	1.059310	0.945492	-0.429074	0.054508	13.6879
8	0.945492	1.074910	-0.429074	0.529655	0.982801	-0.159270	0.017199	3.7962
9	0.982801	1.074910	-0.159270	0.264828	1.004095	0.041718	0.004095	2.1208
10	0.982801	1.004095	-0.159270	0.041718	0.999676	-0.003240	0.000324	0.4421
11	0.999676	1.004095	-0.003240	0.020859	0.999994	-0.000059	0.000006	0.0319
12	0.999994	1.004095	-0.000059	0.010430	1.000006	0.000057	0.000006	0.0012

Final result after 12 iterations: c = 1.000006, Absolute Error = 5.70E-06, Relative Approx. Error = 0.0012%

Summary After 12 Iterations

Method	Root Estimate	Absolute Error	Relative Approx. Error (%)
Bisection	1.000073	0.000073	0.0317
False Position	0.782716	0.217284	5.40
Modified False Position	0.999999	0.000001	0.0006



Ranking of Convergence Speed

$$\text{Modified False Position} > \text{Bisection} \gg \text{False Position}$$

Interpretation

- **Modified False Position** converges the fastest. The error decreases rapidly after only a few iterations and reaches approximately 10^{-6} by iterations 11 to 12.
- **Bisection** converges steadily and reliably. Although it is slower than the Modified False Position method, it consistently reduces the error at each iteration and reaches an error of approximately 10^{-4} after 12 iterations.
- **False Position** converges much more slowly for this function. Even after 32 iterations, its error remains around 1.5×10^{-3}, which is significantly larger than the errors obtained by the other two methods.

Bracketing Method (from textbook problems)

5.3 Determine the real root of $f(x) = -26 + 85x - 91x^2 + 44x^3 - 8x^4 + x^5$:

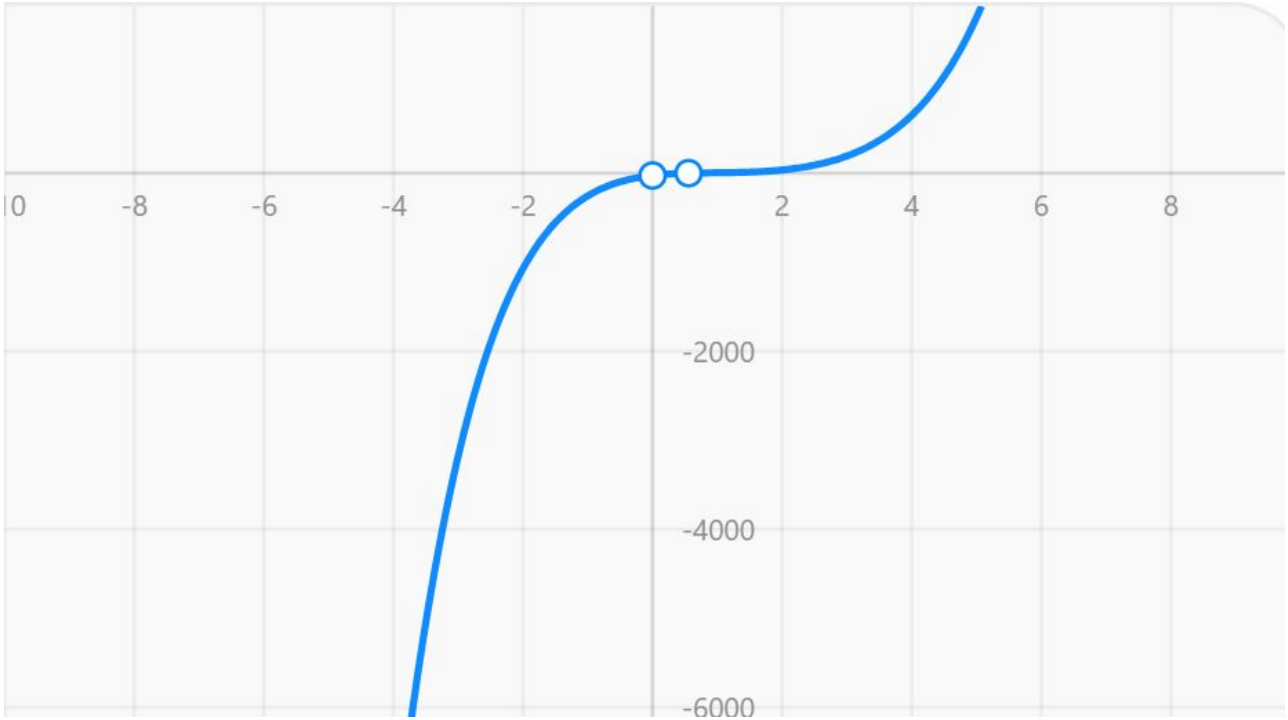
(a) Graphically.

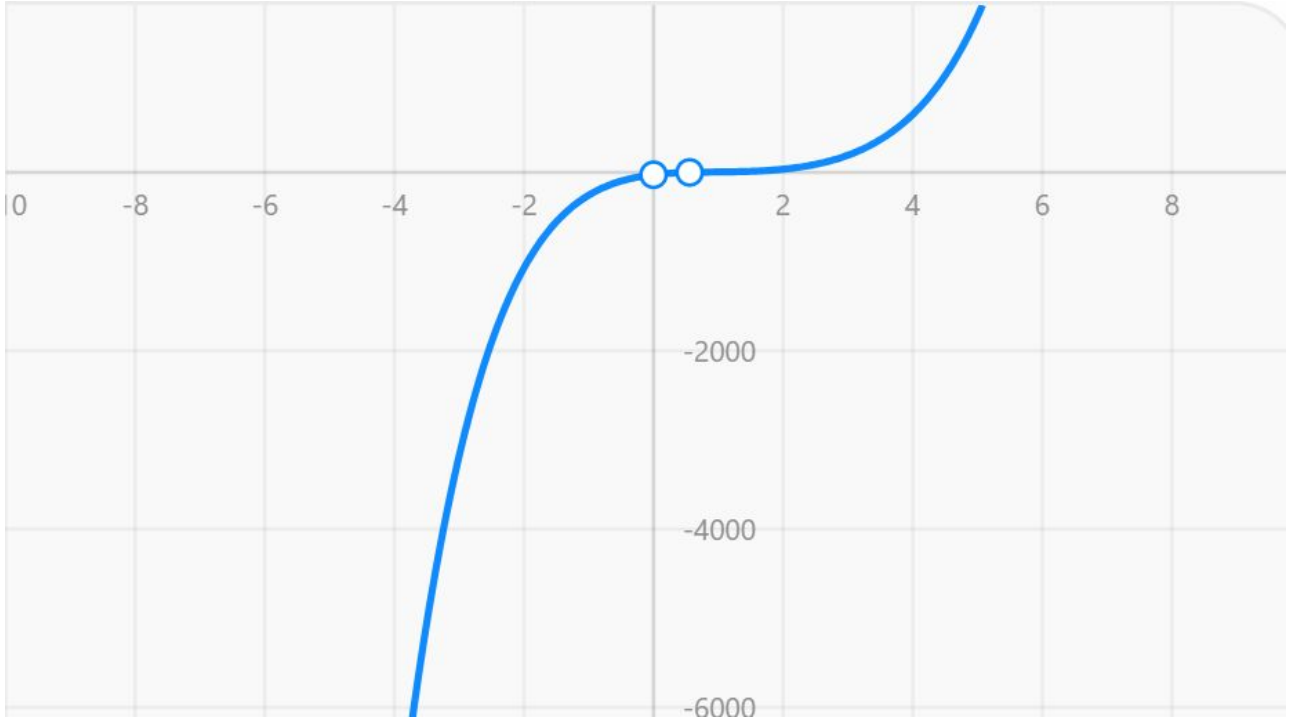
(b) Using bisection to determine the root to $\varepsilon_s = 10\%$. Employ initial guesses of $x_l = 0.5$ and $x_u = 1.0$.

(c) Perform the same computation as in **(b)** but use the false-position method and $\varepsilon_s = 0.2\ \%$.

$$f(x) = x^5 - 8x^4 + 44x^3 - 91x^2 + 85x - 26$$

Graphically





From the graph, the curve crosses the x-axis between **0.5 and 1.0**, closer to **0.6**.

👉 **Graphical estimate:**

$$x \approx 0.6$$

(b) Bisection ($x_l = 0.5, x_u = 1.0, \varepsilon_s = 10\%$)

Initial check: $f(0.5) = -1.219$ and $f(1.0) = 5.0$, so there is a sign change.

Iter	x_l	x_u	x_r	$f(x_r)$	ε_a
1	0.5	1.0	0.75	+2.831	–
2	0.5	0.75	0.625	+1.195	20.0%
3	0.5	0.625	0.5625	+0.106	11.1%
4	0.5	0.5625	0.53125	−0.524	5.88%

$\varepsilon_a = 5.88\% < 10\%$, so the iteration stops.

Root ≈ 0.531 ($\varepsilon_a = 5.88\%$)

(c) False position ($x_l = 0.5, x_u = 1.0, \varepsilon_s = 0.2\%$)

Each estimate uses $x_r = x_u - f(x_u)(x_l - x_u) / [f(x_l) - f(x_u)]$.

Iter	x_l	x_u	x_r	$f(x_r)$	ε_a
1	0.5	1.0	0.59799	+0.7505	–
2	0.5	0.59799	0.56065	+0.0703	6.66%
3	0.5	0.56065	0.55734	+0.0101	0.59%
4	0.5	0.55734	0.55687	≈ 0.001	0.085%

$\varepsilon_a = 0.085\% < 0.2\%$, so the iteration stops.

Root ≈ 0.5569 ($\varepsilon_a = 0.085\%$)

False position converged to a much tighter tolerance (0.2% vs. 10%) in the same 4 iterations. Its estimate of 0.5569 also agrees with the graphical result of about 0.56, whereas bisection's 0.531 is still fairly rough. This is because f is nearly linear near the root, which is where false position works best.